

Vertex AI Gemini Pro Smoke Test (No API Key)

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Goal

Run a minimal Python script on a GCP VM to call Gemini Pro through Vertex AI using Application Default Credentials (ADC). This validates that the VM/service account can access Vertex AI without using an API key.

Pre-requisites

- 1) Vertex AI API enabled in the target project.
- 2) The VM must have an attached service account with permissions to call Vertex AI (commonly *roles/aiplatform.user*).
- 3) Python 3.10+ available on the VM.

Files to create

requirements.txt

```
google-genai>=0.5.0
google-auth>=2.30.0
```

app.py

```
import os
import sys
import google.auth
from google import genai

def must_env(name: str) -> str:
    v = os.getenv(name)
    if not v:
        raise SystemExit(f"Missing env var: {name}")
    return v

def main() -> None:
    # Required
    project_id = must_env("PROJECT_ID")
    # Optional (Vertex regions: e.g., us-central1, europe-west4, etc.)
    location = os.getenv("LOCATION", "us-central1")

    # Application Default Credentials (ADC)
    # On a GCP VM, this comes from the VM's attached service account (no API key needed).
    creds, adc_project = google.auth.default()
    print(f"[auth] Using ADC. adc_project={adc_project!r} target_project={project_id!r}")

    # Vertex AI backend (not Gemini Developer API)
    client = genai.Client(vertexai=True, project=project_id, location=location)

    # Pro model (Vertex AI)
    model = os.getenv("MODEL", "gemini-3-pro")

    prompt = " ".join(sys.argv[1:]).strip()
    if not prompt:
```

```

    prompt = "Reply with: 'Vertex smoke test OK' and the region you're running in."

    print(f"[request] model={model} location={location}")
    resp = client.models.generate_content(model=model, contents=prompt)

    print("\n=== MODEL RESPONSE ===")
    print(resp.text)

if __name__ == "__main__":
    main()

```

Run steps

```

# 1) Create a virtualenv and install deps
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

# 2) Set env vars (no API key required)
export PROJECT_ID="jlr-dl-iqm"
export LOCATION="europe-west2"      # change if needed
export MODEL="gemini-3-pro"          # Pro model on Vertex AI

# 3) Run the prompt
python app.py "Reply with: Vertex smoke test OK"

# Expected: prints [auth] ... and a response from the model.

```

Troubleshooting

Common errors:

- 403 PERMISSION_DENIED:
The VM's attached service account is missing Vertex AI permissions.
Typical role: roles/aiplatform.user (and any org-specific constraints).
- 404 NOT_FOUND or model not recognized:
Ensure the model name is enabled/available in this project/region.
- 429 RESOURCE_EXHAUSTED:
Quota or rate limit reached. Retry or check Vertex AI quotas.

Verification tips (on the VM):

- Confirm the active service account:
curl -H "Metadata-Flavor: Google" \
 http://metadata.google.internal/computeMetadata/v1/instance/service-accounts/default/email